A Narrow Note on the OPH Gravity Chain

Isolating the Jacobson Route

B. Muller

May 17, 2026

Paper release: r1451 **Released:** May 17, 2026

Observer-Patch Holography (OPH) contains a gravity branch written for readers familiar with Jacobson's 1995 and 2016 entanglement-equilibrium arguments. The conditional chain isolated here is

$$\text{observer overlap} \to \text{modular cap geometry} \to \text{null half-sided inclusion} \to T_{kk}$$
$$\to \delta S_{\text{gen}} = 0 \to G_{ab} + \Lambda g_{ab} = 8\pi G\, \langle T_{ab} \rangle.$$

The chain is conditional. In particular, fixed-cap generalized-entropy stationarity, the downstream density-upgrade template used when one wants an explicit local $T_{kk}(v, \Omega)$ density, and the bounded-interval projective step used in the small-ball bridge remain explicit premises rather than derived outputs.

The same observer-overlap axiom set is also used elsewhere in the SM/GR derivation paper to support a conditional Lorentz branch, a schedule-independent overlap-consistency normal form, and a separate compact gauge-reconstruction branch targeting Standard Model structure.

1. Cap Modular Geometry and the Lorentz Branch

OPH starts from a net of patch algebras on a screen S^2, overlap consistency of neighboring observers, a local MaxEnt/refinement-stable branch, and a recoverable generalized entropy functional. The gravity branch then takes a further scaling-limit premise: the refinement-stable cap net lies in the Bisognano–Wichmann geometric modular class.

For each cap $C \subset S^2$, let $\lambda_C(s) \subset \text{Conf}^+(S^2)$ be the standard cap-preserving conformal subgroup, normalized so that the null blow-up near a smooth cut acts by

$$v \mapsto e^{-s} v.$$

If B_C is the infinitesimal generator of that geometric action, the OPH gravity theorem in the SM/GR derivation paper states

$$K_C = 2\pi B_C,$$

where K_C is the cap modular Hamiltonian on the OPH Bisognano–Wichmann branch.

This gives the local kinematic group:

$$\text{Conf}^+(S^2) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1).$$

Hence, once cap modular flow is geometric in the stated scaling branch, the induced local kinematics is the connected Lorentz group.

2. Null Half-Lines and the Borchers Generator

Next one blows up a cap near a smooth entangling cut and restricts to a chosen null generator Ω. Let v be an affine null coordinate with the cut at $v = 0$, and define the half-line algebras

$$H_a := (a, \infty), \qquad \mathcal{M}_a(\Omega) := \overline{\bigvee_{a<c<d<\infty} \mathcal{A}((c,d),\Omega)}.$$

Because the cap modular action is geometric with the 2π normalization, the blow-up modular action on the null coordinate is

$$v \mapsto e^{-2\pi t} v,$$

hence

$$\sigma_t^\omega\big(\mathcal{M}_a(\Omega)\big) = \mathcal{M}_{e^{-2\pi t}a}(\Omega).$$

For $t \le 0$, this implies

$$\sigma_t^\omega\big(\mathcal{M}_a(\Omega)\big) \subseteq \mathcal{M}_a(\Omega),$$

so $\mathcal{M}_a(\Omega) \subset \mathcal{M}_0(\Omega)$ is a half-sided modular inclusion.

At that point Borchers–Wiesbrock applies. One gets a unique strongly continuous unitary group

$$U_\Omega(a) = e^{iaP_\Omega}, \qquad P_\Omega \ge 0,$$

with

$$U_\Omega(a)\, \mathcal{M}_b(\Omega)\, U_\Omega(a)^* = \mathcal{M}_{a+b}(\Omega),$$

and the affine covariance relation

$$\Delta_0(\Omega)^{it} U_\Omega(a) \Delta_0(\Omega)^{-it} = U_\Omega(e^{-2\pi t} a).$$

Differentiating gives

$$[K_0(\Omega), P_\Omega] = -\,i\,2\pi P_\Omega,$$

and for translated half-lines

$$K_a(\Omega) = K_0(\Omega) - 2\pi a\, P_\Omega.$$

In the OPH chain, the positive null-translation generator is obtained from the cap blow-up geometry plus half-sided inclusion on the OPH scaling branch.

3. From the Half-Line Generator to Null Stress

The half-line generator does not by itself give the Einstein equation. In the SM/GR derivation paper, the half-line generator/charge identification is fixed on the stated half-line family. The Jacobson step keeps the density-upgrade template as an explicit premise when one wants the explicit weighted $T_{kk}(v,\Omega)$ integral form, together with the bounded-interval projective transport used in the small-ball bridge:

$$\widetilde{K}[(a,\infty),\Omega] = 2\pi \int_a^\infty (v-a)\, T_{kk}(v,\Omega)\, dv + E^{(\eta)}_{(a,\infty),\Omega}.$$

Here $E^{(\eta)}$ is a carried remainder controlled in matrix elements. The half-line generator/charge identification is fixed by the stated half-line family, so the small-ball/Jacobson step carries transport to bounded null intervals as an explicit premise. The later tensor upgrade is still defined only modulo the null-invisible metric term.

Knowledge of T_{kk} in all null directions then determines the full stress tensor modulo a metric term: if a symmetric tensor X_{ab} obeys

$$X_{ab} k^a k^b = 0$$

for every null vector k, then

$$X_{ab} = \phi g_{ab}$$

for some scalar ϕ.

So the null branch determines T_{ab} up to the usual ϕg_{ab} ambiguity, exactly the ambiguity that later shows up as the cosmological-constant term.

4. The Jacobson Step

Specialize to the physical $d = 4$ scaling regime and consider a small cap C. For admissible first-order state variations at fixed cap, fixed cap size or volume, and fixed conserved charges, OPH uses the first law

$$\delta S_C = \delta\langle K_C\rangle = 2\pi\,\delta\langle B_C\rangle.$$

It also uses the generalized-entropy split

$$S_{\mathrm{gen}}(C) = S_{\mathrm{bulk}}(C) + \frac{A(C)}{4G}.$$

The key extra premise is then stated plainly:

$$\delta S_{\mathrm{gen}}(C) = 0.$$

This is the fixed-cap generalized-entropy stationarity assumption. OPH does *not* claim to derive this from MaxEnt alone.

In the locally Lorentzian $d = 4$ small-ball regime, if $\delta\langle T_{00}\rangle$ is approximately constant across a radius-ℓ ball and the carried remainder contributes only $o(\ell^4)$, the OPH small-ball lemma in the SM/GR derivation paper gives

$$\delta S_{\mathrm{bulk}}(C) = \frac{8\pi^2\ell^4}{15}\,\delta\langle T_{00}\rangle + O(\ell^5\partial T) + o(\ell^4).$$

For a fixed-volume small ball, the standard geometric variation is

$$\delta A\big|_{V,\Lambda_{\mathrm{ref}}} = -\frac{4\pi\ell^4}{15}\,\delta(G_{00} + \Lambda_{\mathrm{ref}}g_{00})\,.$$

Putting these into $\delta S_{\mathrm{gen}} = 0$ gives

$$0 = \frac{8\pi^2\ell^4}{15}\,\delta\langle T_{00}\rangle - \frac{\pi\ell^4}{15G}\,\delta(G_{00} + \Lambda_{\mathrm{ref}}g_{00}) + O(\ell^5\partial T) + o(\ell^4).$$

Dividing by ℓ^4 and taking the small-ball limit yields the rest-frame relation

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\,\delta\langle T_{00}\rangle.$$

Finally, overlap consistency across observers through the same point supplies all local timelike directions. Combined with the null reconstruction lemma above, this upgrades the rest-frame statement to

$$G_{ab} + \Lambda g_{ab} = 8\pi G\,\langle T_{ab}\rangle$$

modulo the isolated metric-term ambiguity.

5. What Is Borrowed and What Is OPH-Specific

Imported mathematical or physical ingredients

- Bisognano–Wichmann geometric modular action and the 2π normalization.
- Borchers–Wiesbrock half-sided modular inclusion and the resulting positive translation generator.
- Jacobson's small-ball first-law or entanglement-equilibrium step from $\delta S_{\mathrm{gen}} = 0$ to the Einstein relation.

What OPH is trying to add

- A concrete observer-patch framework in which caps are primary objects and overlap consistency is the organizing principle.
- A route from cap modular geometry to a null half-line modular pair, with half-sided inclusion derived from cap blow-up geometry.
- An all-directions upgrade driven by overlap consistency across neighboring observers.
- One shared axiom set: in the SM/GR derivation paper the same observer-overlap structure is also used for the Lorentz branch, the overlap-repair normal form, and a separate gauge-reconstruction program aimed at Standard Model structure.
- A dependency ledger that keeps the remaining burdens explicit instead of treating them as settled.

The remaining explicit burdens

- the geometric modular branch assumption on the refinement limit;
- the downstream density upgrade from weak endpoint derivatives to an explicit local null density when one wants the weighted integral form;
- bounded-interval transport of the half-line result through the separate projective branch;
- the fixed-cap stationarity premise $\delta S_{\mathrm{gen}} = 0$.

The resulting conditional Jacobson branch is the mathematical object isolated by this note.

References

1. T. Jacobson, "Thermodynamics of spacetime: The Einstein equation of state," *Phys. Rev. Lett.* 75 (1995), 1260–1263.
2. T. Jacobson, "Entanglement equilibrium and the Einstein equation," *Phys. Rev. Lett.* 116 (2016), 201101.
3. J. J. Bisognano and E. H. Wichmann, "On the duality condition for quantum fields," *J. Math. Phys.* 17 (1976), 303–321.
4. H.-W. Wiesbrock, "Half-sided modular inclusions of von Neumann algebras," *Commun. Math. Phys.* 157 (1993), 83–92.